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A comprehensive review of security vulnerabilities in heavy-duty vehicles: Comparative insights and current research gaps

Narges Rahimi [a],*, Beth-Anne Schuelke-Leech [a], Mitra Mirhassani [b]

[a] Department of Mechanical, Automotive, and Materials Engineering, University of Windsor, Windsor, ON N9B 3P4, Canada
[b] Department of Electrical and Computer Engineering, University of Windsor, Windsor, ON N9B 3P4, Canada

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ABSTRACT

The increasing connectivity and integration of advanced technologies in vehicular systems have amplified the need for robust cybersecurity measures, particularly in heavy-duty (HD) vehicles, which are crucial to commercial transportation. Despite their importance, HD vehicles have received less attention in cybersecurity research compared to light-duty (LD) vehicles, leaving critical vulnerabilities unaddressed. This paper aims to bridge this gap by conducting a thorough analysis of the unique security challenges faced by HD vehicles. By comparing HD vehicles with LD vehicles, we identify distinct and vulnerabilities in two key areas: intra-vehicle networks and external connections. The study includes a comprehensive literature review focused on the cybersecurity of heavy- and medium-duty vehicles, through which we identify prevalent threats and potential mitigation strategies. This analysis underscores the necessity for enhanced protocol security and advocates for a detailed examination of both intra-vehicle networks and external connections.

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* Corresponding author.
E-mail address: rahimi5@uwindsor.ca (N. Rahimi).

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1. Introduction

With the rapid expansion of connected and intelligent vehicle technologies, the importance of robust cybersecurity has become increasingly critical. As vehicles become more interconnected, they are exposed to a broader range of potential threats, underscoring the need for effective security measures. Concurrently, technological advancements lead to the rapid development and integration of new features in modern vehicles, which enhances their capabilities but also increases their complexity and potential vulnerabilities (Pan et al., 2017). This dual progression necessitates prioritizing cybersecurity to protect the expanding digital infrastructure of modern vehicles. In fact, while connected and intelligent vehicles offer notable advancements in convenience and functionality, they also introduce new risks that can be exploited by malicious actors. This is especially true for heavy-duty vehicles, which are integral to the logistics and transportation sectors. The potential impact of a cyber attack on these vehicles could be devastating, affecting not only the safety of the vehicle occupants but also causing widespread disruptions in supply chains and economic activities. Despite their importance, considerable academic attention in the realm of automotive cybersecurity research has been focused primarily on addressing safety and security issues in light-duty and regular vehicles since 2010. While there is substantial work on the security of heavy-duty vehicles, these efforts remain insufficient to address the full range of vulnerabilities, primarily due to limited funding and the high costs associated with experimental research in this area (National Motor Freight Traffic Association, 2015). Heavy-duty vehicles, characterized by a gross vehicle weight rating (GVWR) exceeding 26,000 pounds, include trucks and buses designed for robust freight transport, while medium-duty vehicles, falling between light-duty and heavy-duty categories, encompass delivery vans and larger commercial vehicles. The distinct operational demands and technological architectures of heavy and medium duty vehicles pose unique challenges, necessitating specialized cybersecurity considerations. This survey paper aims to investigate the security vulnerabilities of heavy-duty vehicles, highlighting their differences from light-duty vehicles and passenger cars.

The remainder of the study is organized into five main sections: an overview of heavy-duty vehicles, a comparison of security vulnerabilities between heavy-duty and light-duty vehicles, a review of existing literature and research from 2010 to 2024 in HD vehicle cybersecurity, identification of research gaps and implications for future research, and a concluding summary of findings.

2. Background

The Gross Vehicle Weight Rating (GVWR) is a classification system released by the Federal Highway Administration of the United States (Federal Highway Administration, 2024). This system broadly categorizes vehicles into three groups: light-duty vehicles (LD) for Classes 1 and 2, medium-duty vehicles (MD) for Classes 3 to 6, and heavy-duty vehicles (HD) for Classes 7 and 8, as illustrated in Table 1. Heavy-duty vehicles include city transit buses, school buses, tour buses, heavy trucks, fire trucks, fuel trucks, cement mixers, and refrigerator vans. Commercial vehicles, especially trucks, are vital components in logistics, playing a crucial role in economic prosperity. As indicated by D'Anna (2018), the American economy heavily depends on commercial vehicles, particularly trucks. For instance, in the U.S., 80 percent of goods transport relies on trucks, and any disruption in truck traffic results in rapid economic instability. As highlighted by American Trucking Associations (2023), National Motor Freight Traffic Association (NMFTA), when trucks stop, America stops. In addition to the freight system, public transportation is considered the "lifeblood" of cities and is vital for the economic, environmental, social, and overall well-being of urban areas. It supports sustainable and inclusive growth while enhancing the quality of life for residents. Public transportation reduces traffic congestion, lowers emissions, and provides accessible mobility options, contributing to the overall resilience and efficiency of urban environments (Miller et al., 2016). Despite the critical role of these vehicles in economic stability, sustainability, and logistics, and the numerous stakeholders affected by security threats, publicly available research on the security of HD vehicles is not sufficient and needs more attention. While notable advancements in the security of HD vehicles have been made in recent years, particularly within Original Equipment Manufacturers (OEMs), there is a lack of comprehensive literature reviews addressing this context.

One of the most significant contributions to the cybersecurity of HD vehicles has been made by the NMFTA, which provided classified information regarding the security of HD vehicles based on NIST (National Institute of Standards and Technology) guidelines (NMFTA, 2025). To the best of our knowledge, the only survey publication reviewing the security of HD vehicles is the report released by the NMFTA (National Motor Freight Traffic Association) in 2015, which primarily examines the structural aspects of HD vehicles from a security perspective (National Motor Freight Traffic Association, 2015a). As highlighted by NMFTA in their 2015 survey report, additional data and information specific to heavy vehicles exists, though much of it may remain unpublished or otherwise unidentified due to the limited timeframe available for preparing this report (National Motor Freight Traffic Association, 2015b). Therefore, the aim of this survey is to provide an updated review of HD vehicle security in literature. It should be noted that numerous publications and studies have focused on the security of passenger cars and LD vehicles, as many security considerations and structural aspects are applicable to both HD and LD vehicles. Identifying the differences between HD and LD vehicles in terms of security vulnerabilities is essential for uncovering the specific security challenges unique to HD vehicles, which is addressed as a first contribution in this study.

3. Comparative study of HD and LD vehicles

Cyber–Physical Systems (CPS) are integrated systems in which computational devices closely interact with the physical environment (Furrer, 2022). These systems combine computing, networking, and physical processes, with embedded computers and networks monitoring and controlling physical processes. Often, there are feedback loops where physical processes influence computations and vice versa. Modern and connected vehicles, as noted by Humayed et al. (2017), exemplify CPS, where computational devices closely interact with the physical environment. Connected vehicles include key components such as the Powertrain (PTC), Chassis and Safety (CSC), Body Electronics (BEM), and Head Unit (HU) (Henniger et al., 2009), which communicate with each other and with infrastructure. These vehicles rely on embedded systems to manage engine performance, safety features, and infotainment, illustrating the complex interplay between computational and physical elements. While the connectivity and intelligence of these vehicles enhance their capabilities, making transportation systems more efficient, safe, and reliable, this increased connectivity also expands the attack surface for cyber threats. Consequently, critical systems are exposed to potential hacking, data breaches, and malicious attacks that can compromise vehicle safety and functionality.

The expansion of connected vehicle technology extends not only to passenger cars but also to HD vehicles. Today, most HD vehicles

Table 1
Vehicle type classification (U.S. Department of Energy, 2018).

Class	Weight range	Vehicles name
1	Less than 6 k lbs. (light)	Full Size Pickup, Mini Pick up, Minivan, SUV, Utility Van
2	6 K to 10 K lbs. (light)	Crew Size pickup, Full Size Pickup, Mini Bus, Mini Van, Step Van, Utility Van
3	10 K to 14 K lbs. (medium)	City Delivery, Mini Bus, Walk In
4	14K to 16K lbs. (medium)	City Delivery, Conventional Van, Landscape Utility, Large Walk In
5	16K to 19.5K lbs. (medium)	Bucket, City Delivery, Large walk In
6	19.5K to 26K lbs. (medium)	Beverage, Rack, School Bus, Single Axle Van, Stake Body
7	26K to 33K lbs. (heavy)	City Transit Bus, Furniture, Truck Tractor and Semi-Trailer, Home Fuel, Medium Semi Tractor, Refuse, Tow
8	Greater than 33K lbs. (heavy)	Cement Mixer, Dump, Fire Truck, Fuel, Heavy Semi Tractor, Refrigerated Van, Semi Sleeper, Tour Bus

are equipped with advanced communication and fleet management systems. To assess the security of HD vehicles effectively, it is crucial to examine their systems and understand how they differ from LD vehicles. Evaluating these differences will help determine whether the same cybersecurity approaches, vulnerability analyses, and threat management strategies used for LD vehicles are sufficient or if HD vehicles require more robust cybersecurity measures and tailored precautionary implementations.

To explore these differences, it is important to consider and address the most critical attack surfaces and vulnerabilities specific to vehicular systems in HD vehicles. Connection points are often seen as the primary attack surface in cybersecurity, representing the initial vulnerabilities in a system's security. This study focuses on the differences between HD and LD vehicles with respect to two main types of connections: intra-vehicle and external vehicle connections. Intra-vehicle connections refer to the internal communication systems within a single vehicle (Pan et al., 2017), while external vehicle connections involve communication between the vehicle and external entities such as servers, cloud services, or other vehicles. By examining these connection types, the study aims to identify and understand potential security vulnerabilities unique to HD vehicles.

It is important to note that many vulnerabilities and attack surfaces identified in HD vehicles are also applicable to MD vehicles. Furthermore, MD vehicles, depending on their model and type, may share similarities with LD vehicles. However, this study focuses primarily on general HD vehicles and does not explore MD vehicles, specific model types, or brands.

3.1. Intra-vehicle connections

The Electronic Control Unit (ECU) is a vital component in the vehicular system, responsible for managing and controlling various aspects of the vehicle operation. The embedded system in vehicles consists of various ECUs, each group having a specific function, and the complexity and type of vehicle determine the number and type of ECUs present. Modern passenger vehicles are equipped with multiple ECUs responsible for various functions, such as Adaptive Cruise Control (ACC), Anti-Lock Brake System (ABS), Battery Management System (BMS), Engine Control Module (ECM), Light Switch Module (LSM), Park Distance Control (PDC), and others. Typically, these vehicles are equipped with a varying number of ECUs, depending on their complexity and features, each dedicated to managing specific systems and contributing to the overall functionality, safety, and performance of the vehicle (Salunkhe et al., 2016). In contrast, HD vehicles, tailored to specific applications, may share similar ECUs but with distinct architectures (Wolf and Lambert, 2017). Engineered for tasks like freight transportation and construction, these vehicles prioritize efficiency and compliance (Weimerskirch and Gaynier, 2015). Common ECUs in HD vehicles include the Engine Control Unit (ECU), Transmission Control Unit (TCU), and Vehicle Control Unit (VCU). In certain electric vehicle configurations, additional ECUs may be deployed to manage the intricacies of the battery system.

The roles of these ECUs are pivotal, overseeing the functions of the engine, transmission, and overall vehicle control. For example, the Bendix EC-60 brake ECU, designed for commercial and HD/MD vehicles, integrates features such as Anti-Lock Braking System (ABS) and electronic stability control (ESC). This ECU primarily uses kinematic sensors, such as wheel speed, to monitor and control braking performance in real-time, enabling informed decisions for optimizing braking functionality (Burakova et al., 2016). It is noteworthy that ECUs, depending on their functionality, are tailored distinctively for HD and MD vehicles. This customization ensures that these vehicles meet the specific demands of their applications while adhering to industry standards and safety requirements. Overall, while both passenger and HD/MD vehicles utilize ECUs to control various functions, the complexity, number, and architecture of ECUs can vary depending on the type of vehicle and its intended application. Understanding the differences in ECU systems between these vehicle types is crucial for evaluating their respective cybersecurity measures and vulnerabilities (Ford Company).

In the context of embedded systems in vehicles, a protocol is a set of rules and conventions used for communication between different ECUs. The protocols ensure that data is transmitted efficiently, reliably, and correctly across the vehicle's network (Salunkhe et al., 2016; Wolf and Lambert, 2017). Automotive systems can employ various communication protocols. For instance, communication protocols such as CAN (Controller Area Network), MOST (Media Oriented Systems Transport), Ethernet, FlexRay, VAN (Vehicle Area Network), and LIN (Local Interconnect Network) have been implemented (Hegde et al., 2013). The most widely used protocol in regular vehicles is the CAN bus, while HD vehicles typically use the J1939 protocol (Johansson, 2012), which is based on CAN but includes distinct features that will be discussed further. The most significant security differences between HD and LD vehicles emerge from their use of different protocols (Mukherjee et al., 2016). Note that a comprehensive discussion of all communication protocols and their characteristics is beyond the scope of this research. In this study, CAN has been identified as the primary communication bus for passenger vehicles, while J1939 is designated for HD vehicles. The term "CAN bus" stands for Controller Area Network bus. Its development began at Robert Bosch GmbH in 1983, and it was included in the ISO (International Organization for Standardization) standard in 1993 under ISO11898 (Gay, 2024) as illustrated in Fig. 1. The primary aim of CAN was to create a networking

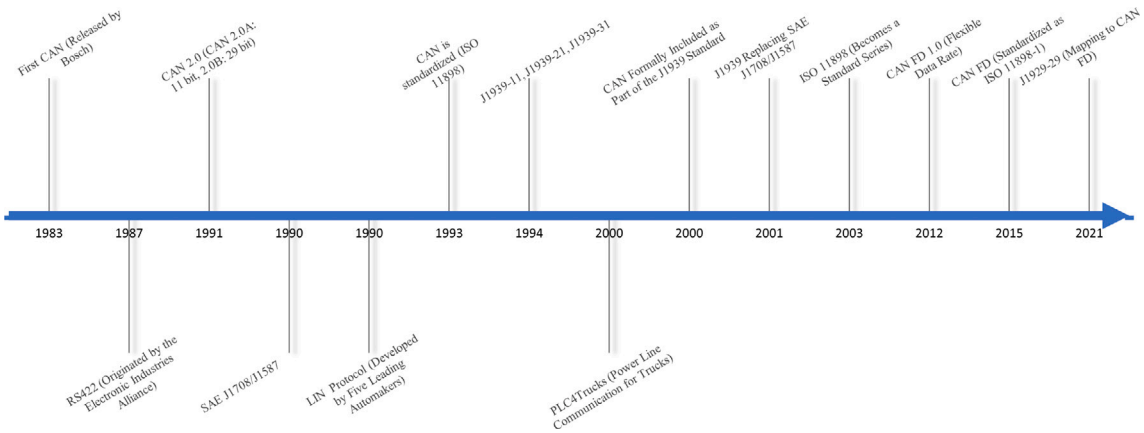

Fig. 1. Development Timeline of CAN, J1939, and Other Communication Protocols Used for HD Vehicles (C.S.S. Electronics, 2024; Wang et al., 2012; Khanapurkar et al., 2008; National Motor Freight Traffic Association (NMFTA), 2025).

system connecting various electronic modules in vehicular systems, thus eliminating the need for a dedicated computer (Salunkhe et al., 2016; Pan et al., 2017). Conversely, HD vehicles predominantly employ J1939 (Mukherjee, 2023). This standard finds wide adoption across various industries utilizing diesel engine vehicles, including "On-Road" (Truck and Trailer) and "Off-Road" (Construction, Cranes, etc.) applications. J1939 is built upon two older standards defined by SAE, J1708 and J1587. Additionally, protocols for HD/MD vehicles based on J1939 have been developed, such as MilCAN (military applications), ISOBUS (agricultural industry), and NMEA 2000 (marine applications) (Prasad et al., 2019). In this comparison, "CAN" refers to the protocol used in LD vehicles and passenger cars, while J1939 is built upon CAN. To highlight the differences between the two, the characteristics and concepts of these protocols can be discussed as follows:

1. The first metric in protocol is topology, which defines the structure and shape of connections between ECUs. Various topologies have been employed in vehicle networks, including star, point-to-point, chain, and bus (Mukherjee, 2023; Hegde et al., 2013; Johansson, 2012). Among these, the simplest and most widespread topology in vehicles is the "bus" topology. A bus topology refers to a network design where multiple ECUs are connected along a single communication line or bus (Hegde et al., 2013; Johansson, 2012). This setup allows the ECUs to communicate with each other by transmitting data over the shared bus. Both protocols J1939 and CAN employ bus topology (Johansson, 2012).

2. The second aspect of a protocol is OSI (Open Systems Interconnection) model, which provides a framework for standardizing communication protocols. The OSI model is a conceptual framework used to categorize communication functions into seven distinct layers (Zemrowski, 1994). Each layer serves specific functions and interacts with the layers directly above and below it. Understanding how vehicle communication protocols fit into this model helps in designing and troubleshooting automotive networks. These seven layers consist of Physical Layer, Data Link Layer, Network Layer, Transport Layer, Session Layer, Presentation Layer and Application Layer (Zemrowski, 1994). While the CAN bus utilizes only the Physical and Data Link layers of the OSI model, J1939 incorporates the Application, and Network layers in addition to the Physical and Data Link layers (Zhang and Lu, 2020; Daily et al., 2021), as indicated in Fig. 2. In addition to these seven layers, J1939 also includes two more layers: the Network Management layer (Layer 8), which manages communication tasks such as initialization, message prioritization, and fault detection, and the Network Security layer (Layer 9), which ensures communication integrity through security measures like encryption, access control, and authentication (CyberTruck Challenge, 2025).

3. In the physical structure of the CAN bus, two wires are used, including high and low voltage ones. Two wire structures in CAN include single parallel differential pairs and twisted differential pairs. The twisted form is generally preferred for its ability to reduce electromagnetic interference between wires, making the parallel structure less commonly used. Additionally, the bit rate used between two nodes defines the maximum distance of wires, such as 1 Mb/s (40 m), 500

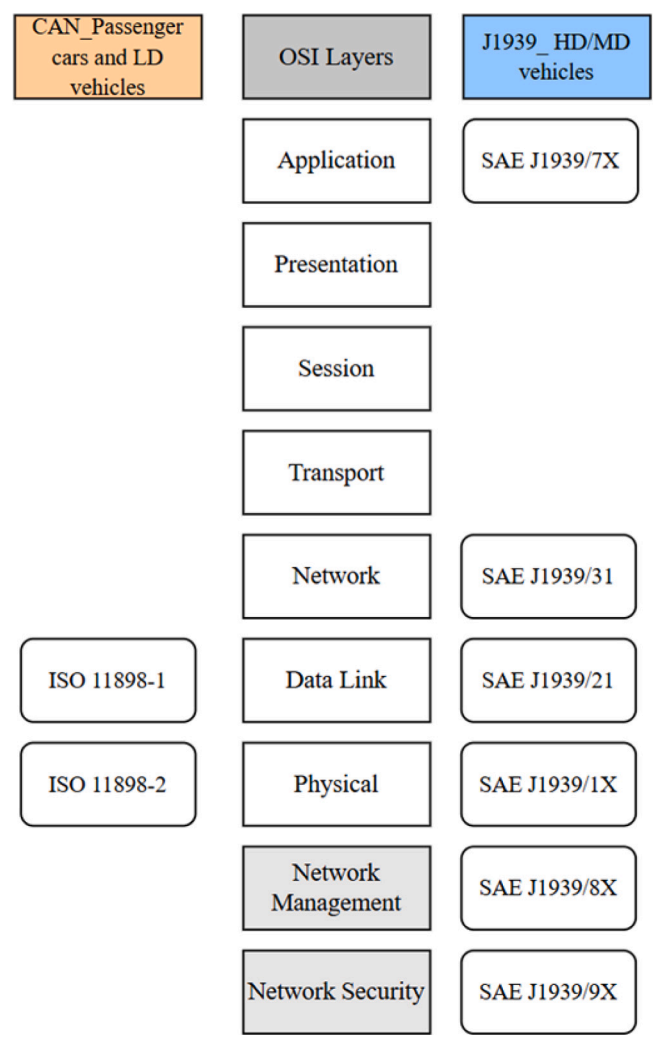

Fig. 2. OSI layers in CAN and J1939 and their associated standard protocols (Mukherjee, 2023; CyberTruck Challenge, 2025).

kb/s (130 m), and 250 kb/s (270 m). On the other hand, in the physical layer of J1939, twisted pair wires are employed, with a bit transfer rate of up to 250 kb/s (a 500 kb/s version is also defined), and a maximum of 30 nodes can be connected to the same bus (Mukherjee, 2023).

4. Another difference between J1939 and the CAN protocol used in passenger and LD vehicles lies in their frame structures. In both protocols, the basic unit of communication is a frame. According to the data link layer, communication over the CAN bus occurs through CAN frames, which consist of several components that facilitate data transmission between ECUs within a vehicle (C.S.S. Electronics, 2024). Both CAN and J1939 systems employ a very similar frame structure. In fact, the J1939 frame is essentially built upon the CAN bus frame, consisting of two main parts: a standard identifier and a data field, controllable by the host (Burakova et al., 2016; Stachowski et al., 2019). The identifier is crucial for efficient communication between nodes or ECUs within the network. It ensures messages are properly identified, enabling an efficient communication system in complex automotive and industrial applications. In the first generation of CAN (CAN 2.0A), the specification allowed up to 11 bits for the identifier and 8 bytes for the data field. The next generation, known as CAN 2.0B and CAN FD, extended the ID to 29 bits and the data to 64 bytes. J1939, on the other hand, utilizes the 29-bit ID, divided into different parts: priority, Parameter Group Number (PGN), source address, and destination address based on the PDU (Protocol Data Unit) format message. This PGN structure allows J1939 to handle a larger data transfer load (Mukherjee, 2023; Stachowski et al., 2019). Fig. 3 illustrates the frame structure in J1939.

5. Another difference is the communication method employed by each protocol. CAN operates as a broadcast protocol, meaning that every node (ECU) on the CAN bus receives all transmitted messages, without built-in support for unicast communication. In other words, each node can observe messages sent by any other node on the network. However, J1939 enhances this framework by introducing mechanisms that enable point-to-point message transfers. Specifically, J1939 incorporates source and destination address fields within its CAN frames to define both the sender and the intended recipient of a message (Mukherjee et al., 2016). However, it is important to note that point-to-point in J1939 is a logical construct—all ECUs on the network still see all messages, and the recipient ECUs filter and process only those messages addressed to them (Mukherjee et al., 2016; CyberTruck Challenge, 2025).

To explore the mechanisms of data transmission in greater detail, Chatterjee et al. (2023) offers valuable insights into the methods utilized in J1939, highlighting the distinctions between Destination-Specific Transfer and Broadcast Transfer. In Destination-Specific Transfer, the sender initiates communication with a Request to Send (RTS) message containing data details, followed by a Clear to Send (CTS) response from the receiver to manage flow control and retransmissions. The sender then transmits Data Transfer (DT) messages, each tagged with a sequence number for reassembly, concluding with an End of Message Acknowledgment (EoMA) from the receiver. In Broadcast Transfer, communication begins with a Broadcast Announcement Message (BAM) detailing the data, followed by DT messages. Unlike Destination-Specific Transfer, Broadcast Transfer does not require an acknowledgment upon completion.

6. J1939 is an open-source protocol, and while open-source protocols are preferred for commercial and HD vehicles due to their user-friendliness, common platform usage, greater adaptability, and performance (Lippert, 2024), they are generally more vulnerable to attacks. However, in the case of J1939, vulnerabilities are reduced due to the incorporation of proprietary messaging and security mechanisms, such as ISO 15765 (International Organization for Standardization, 2023), used by individual manufacturers. These proprietary elements make reverse engineering more difficult and reduce the risk of widespread attacks across HD vehicles. While J1939's standardization could potentially affect multiple vehicles (D'Anna, 2018), the use of proprietary features significantly limits this risk (Shirazi et al., 2020). On the other hand, CAN is a closed-source protocol, so attacking passenger cars often requires modifying hacking techniques based on the specific proprietary protocol used.

Table 2
Comparison of CAN and J1939 Protocols.

Category	CAN	J1939
Physical Layer	Parallel pairs, Twisted pair wires	Twisted pair wires
Transfer Rate	1 Mb/s, 500 kb/s, 250 kb/s	1 Mb/s, 500 kb/s
Length	40 to 270 m	40 to 200 m
OSI Layers	Physical layer, Data link layer	Application Layer, Transport Layer, Network Layer, Physical layer, Data link layer
Frame	2.0A ID: 11 Data: 8 2.0B ID: 29 Data: 8 FD ID: 29 Data: 64	2.0B ID: 29 (PGN + priority + source address)
Protocol	Close-source	Open-source
Architecture	Integrated architecture	Separated architecture: trailer tractor

7. The electronic architecture refers to the way in which ECUs are interconnected and communicate with each other. Many architectures based on brands, years, and models have been released for LD and MD/HD vehicles. However, it should be noted that many HD vehicles, specifically those with a tractor and trailer, have separate parts comprising two main segments: the trailer segment and the tractor segment. The architecture associated with HD vehicles of this type is referred to as a (multi-)flat J1939 architecture with openly published message semantics. On the other hand, the architecture associated with LD and passenger cars is usually known as a (multi-)flat CAN vehicle architecture with proprietary CAN message semantics (Stachowski et al., 2019). In these structures, a central gateway plays a crucial role as the hub of the vehicular architecture, connecting the On-Board Diagnostics (OBD), infotainment, telematics, and ECUs through the CAN network (Stachowski et al., 2019).

In terms of security, increased connectivity and a more distributed architecture make vehicular systems more vulnerable; essentially, more connections create a larger attack surface. An example of the separated architecture of HD vehicles and the central gateway in LD vehicles is illustrated in Fig. 4. It should be noted that many other architectures have been released for different vehicle types, and Fig. 4 is only one example.

The comparison between CAN and J1939 is summarized in Table 2.

3.2. External connection

One of the most critical components in vehicular systems is their interfaces and communication ports. Every cyberattack relies on an entry point, which is a vulnerability or gap in a system's defenses that an attacker exploits to gain access and achieve their goals. Identifying these potential entry points is crucial for assessing the attacker's required effort. A significant difference between HD and LD vehicles lies in their communication interfaces. Exploring these interfaces is essential for a comprehensive comparison of HD and LD vehicles from a cybersecurity perspective. The following discussion, based on the National Highway Traffic Safety Administration (NHTSA) reports (Stachowski et al., 2019; Federal Motor Carrier Safety Administration, 2020), defines both wired and wireless short-range and long-range interfaces and highlights the critical concepts and differences of HD vehicles in this context.

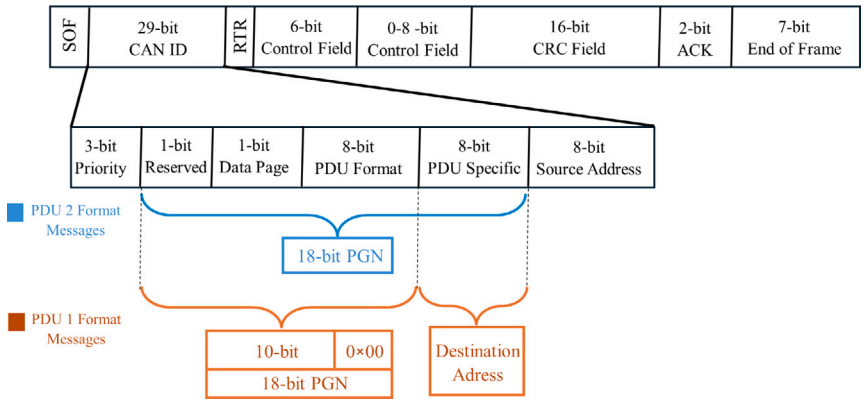

Fig. 3. Frame structure in J1939 (Burakova et al., 2016; CyberTruck Challenge, 2025).

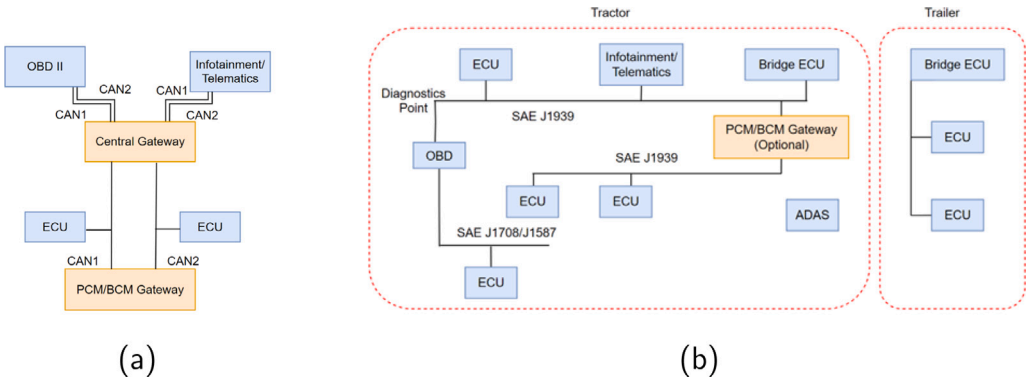

Fig. 4. An example of separated architecture in HD vehicles (b) versus central gateway architecture in LD vehicles (a). Stachowski et al. (2019).

3.2.1. Wired threat vectors

Diagnostic connectors (OBD, ELD, etc.) Both HD and LD vehicles utilize standardized diagnostic connectors to communicate with a vehicle's internal computer systems. These connectors provide access to ECUs for maintenance and repair by technicians (Stachowski et al., 2019; Federal Motor Carrier Safety Administration, 2020). The most widely used diagnostic connector is the On-Board Diagnostics (OBD) port, allowing access to a vehicle's onboard computer systems to retrieve diagnostic trouble codes (DTCs) and real-time sensor data (McCord, 2011). While the NHTSA indicates no difference in OBD technology between HD and LD vehicles (Stachowski et al., 2019), there are key considerations that make HD vehicles more prominent in terms of security concerns. In vehicular systems, OBD connectors serve as gateways to the CAN bus and J1939, presenting a potential vulnerability that attackers may exploit (McCord, 2011). Unlike some modern LD vehicles with secured CAN bus interfaces from OBD connectors, HD vehicles often lack this isolation (Stachowski et al., 2019; McCord, 2011). Additionally, third-party companies offer OBD-based telematics dongles, allowing access to ECUs and monitoring information like speed, engine RPM, and throttle position. Many of these dongles can transmit data wirelessly to fleet management systems, which can be more vulnerable compared to solely wired connections. While both HD and LD vehicles can utilize OBD dongles, they are more commonly used in HD vehicles due to their commercial applications.

In addition to onboard diagnostic interfaces, such as the SAE J1962 port used in LD vehicles (S.A.E. International, 2016), off-board diagnostic connections are utilized in HD vehicles. Interfaces like RP1226 (S.A.E. International, 2024b) and J1939-13 (S.A.E. International, 2024a) are commonly employed in HD vehicles to facilitate communication with the vehicle's ECUs, enabling external systems to monitor and collect data for purposes such as maintenance, fleet management, and diagnostics.

In many commercial vehicles, Electronic Logging Devices (ELDs) connect to intra-vehicle networks, such as J1939, to monitor critical data like engine hours, vehicle motion, and distance traveled (Jepson et al., 2024a). These devices are subject to different regulations depending on the region, including standards set by the United States (Federal Motor Carrier Safety Administration, 2025) and Canada (Transport Canada, 2022). Considering the operational time of HD vehicles and their critical role in commercial activities, their monitoring and maintenance are more crucial than those of LD vehicles (Hedvall, 2016). As a result, diagnostic interfaces are used more frequently in HD vehicles for ongoing monitoring and maintenance.

In addition to the diagnostic connections defined, the NHTSA report indicates that there is no significant difference between the diagnostic tools used in HD vehicles and LD vehicles (Stachowski et al., 2019).

USB ports, CD drives, SD cards, and auxiliary audio inputs Similar USB ports, CD drives, SD card slots, and auxiliary audio jacks are commonly integrated into vehicle infotainment systems in both HD and LD vehicles. These components connect to the vehicle's CAN bus (Jeong et al., 2023). While technology and structure of infotainment modules might be similar, some considerations exist for HD vehicles. Unlike LD vehicles primarily focused on entertainment, HD vehicles may utilize these functionalities for additional purposes like vehicle monitoring. This can necessitate a connection to the critical J1939 bus, introducing a potential security concern if the infotainment system is compromised (Stachowski et al., 2019).

12-Volt accessory outlet Regular vehicles utilize 12-volt accessory outlets solely for providing power to various electronic devices. However, in HD vehicles, these outlets can share wiring with a communication system (PLC) used for data exchange between the truck and trailer. This overlap creates a potential vulnerability because someone

could exploit the power outlet to disrupt communication within the vehicle (Stachowski et al., 2019).

Body builder interface Absent in passenger cars, HD trucks offer a "body builder interface" - a specialized port for connecting add-on equipment directly to the J1939 network, enabling functionalities specific to commercial vehicles (Volvo Truck). In contrast, passenger vehicles do not require such open interfaces since they are not typically modified extensively. While this functionality facilitates customization, it creates a unique attack point if not properly secured. It is worth noting that in HD vehicles, there may be exterior access points, like between a tractor and trailer, which could potentially be vulnerable to unauthorized access (D'Anna, 2018; Stachowski et al., 2019). However, such external access points are generally absent in passenger vehicles, which may give the impression of enhanced security. Nonetheless, attackers can target interior interfaces through alternative methods specific to passenger car designs. These structural distinctions underscore the importance of adopting tailored cybersecurity measures that account for the differing vulnerabilities and attack feasibility associated with each vehicle type.

Trailer power line communication (PLC-J2497) PLC, as a unique connector in HD vehicles, especially class 7 and 8, has been employed to monitor the ABS status of trailers connected to tractors. Regular vehicles do not utilize this connection, making this vulnerability specific to HD vehicles (Stachowski et al., 2019).

3.2.2. Wireless (short-range) threat vectors

Bluetooth Bluetooth, widely utilized in vehicular systems, enables the pairing of mobile phones with infotainment systems and potentially facilitates software updates. This technology is almost well-established and secure (Chen and Chen, 2005), with no significant differences in implementation between HD and LD vehicles (Stachowski et al., 2019). However, the use of Bluetooth-enabled dongles and add-in connections to diagnostic ports may introduce additional vulnerabilities in HD vehicles.

Tire pressure monitoring systems (TPMS) By monitoring tire pressure, this connection can alert drivers by interfacing with CAN and J1939 protocols. Additionally, there are no significant differences between the TPMS technology used in HD and LD vehicles (Stachowski et al., 2019).

Wi-Fi Wi-Fi in automotive systems connects a vehicle's infotainment system to Wi-Fi-enabled devices, enabling activities like streaming and internet access. However, this connectivity poses security risks, potentially allowing cyberattacks through external networks. Vulnerabilities arise from connections to the CAN bus in LD vehicles and the J1939 in HD vehicles, which attackers might exploit to manipulate vehicle functions or access sensitive data (Josephlal and Adepu, 2019). While both vehicle types use Wi-Fi, HD vehicles primarily use it for diagnostics and fleet management, introducing further security considerations (Stachowski et al., 2019; Goel, 2008).

Dedicated short-range communications (DSRC) Dedicated Short-Range Communication (DSRC) technology is being developed by passenger vehicle manufacturers to improve collision prevention through vehicle-to-vehicle and vehicle-to-infrastructure communication. This technology, also being researched in the HD vehicles sector, shows no significant security differences between LD and HD vehicle implementations (Stachowski et al., 2019).

3.2.3. Wireless (long range) threat vectors

GSM/CDMA (telematics) GSM (Global System for Mobile Communications) and CDMA (Code Division Multiple Access) are key standards for digital cellular networks, with GSM recognized as the global standard for mobile communications since 2014. Both LD and HD vehicles use these standards in their infotainment and telematics systems, which often feature cellular modems and antennas for data transmission (Johansson, 2012; Goel, 2008). Advances in 3G/4G cellular technology have enhanced security through authentication and over-the-air encryption. However, these advancements also introduce potential vulnerabilities, especially when integrated into infotainment systems, which could expose the vehicle's CAN bus (in LD vehicles) or J1939 bus (in HD vehicles) to threats. In HD vehicles, the extensive use of telematics for fleet management increases the attack surface, making them more susceptible to exploitation by malicious actors (Stachowski et al., 2019).

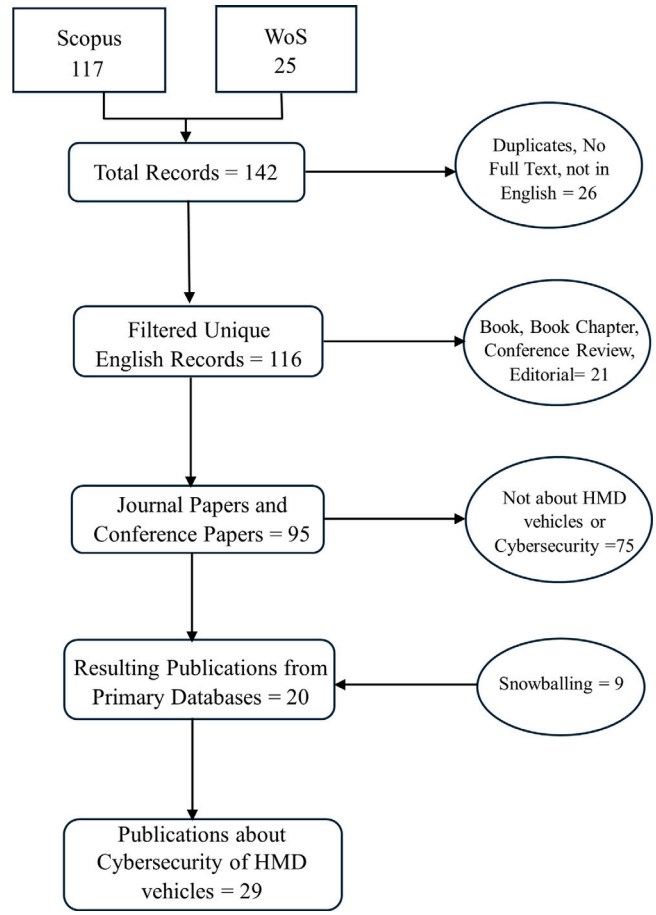

Fig. 5. Overview of Selected Papers and Classification Framework.

4. Review of cybersecurity research in HD/MD vehicles (2010–2024)

The main objective of this section is to provide a structured review of cybersecurity research on HD/MD vehicles from 2010 to 2024, identifying research gaps and potential areas for further investigation. The following steps outline the methodology used:

Database Selection Scopus and Web of Science were selected as the primary digital libraries for this literature review. Scopus was chosen for its extensive collection of relevant articles, while Web of Science provided additional comprehensive coverage and reputable sources.

Search Terms To find the most relevant papers, key terms were defined and divided into two categories:

Cybersecurity: Articles containing variations of the term "cybersecurity" (including "cyber-security") were included to ensure comprehensive coverage. Vehicle Types: The search included terms such

as "Medium Duty Vehicle," "Medium Vehicle," "Medium-duty Vehicle," "Heavy Duty Vehicle," "Heavy Vehicle," "Commercial Vehicle," "truck", "shuttle", "tractor", "trailer", "Mini Bus", "Vintage Bus", "Coach Bus", "Transit Bus", "School Bus", "Intercity Bus", "Shuttle Bus", "Tour Bus", "Electric Bus", "Hybrid Bus", "Double Decker Bus", "Articulated Bus", "Minibus", "Rapid Transit Bus", "Airport Shuttle Bus", "City Bus", "Sightseeing Bus", "Trolleybus", "Autonomous Bus", "Community Bus", and "Luxury Bus".

Search procedure The initial group of articles was filtered based on exclusion criteria. Abstracts of the remaining articles were skimmed according to inclusion criteria to determine their relevance to the research topic.

Selection criteria The scope of this review is limited to articles published between January 2010 and May 2024 that address the cybersecurity of heavy or medium-duty vehicles. The inclusion and exclusion criteria are as follows:

Inclusion criteria:

1. Articles must focus on the cybersecurity of heavy or medium-duty vehicles.
2. Articles must be relevant to the defined search terms.

Exclusion criteria:

1. Articles not written in English.
2. Books, book chapters, conference reviews, or editorials.
3. Articles without full text availability.
4. Duplicate articles, including those found in both Scopus and Web of Science.

Fig. 5 illustrates the analysis of the reviewed papers, highlighting the inclusion and exclusion criteria.

Snowballing To enhance the comprehensiveness of this literature review, additional articles were included through a snowballing technique, which involved reviewing references of initially selected papers to find more relevant studies.

Finally, we have reviewed 29 papers exploring vulnerabilities, cyber-attacks, threat analysis, or risk assessment of heavy-duty or medium-duty vehicles. To provide a better taxonomy, we have separated the papers into two categories: medium-duty vehicles and heavy-duty vehicles. In the literature, only three papers focused on medium-duty vehicles, while the majority examined heavy-duty vehicles, often referred to as heavy or commercial vehicles. Additionally, two papers investigated electric buses, and three publications explored specific heavy-duty vehicles, including construction, agricultural and mining vehicles, as illustrated in Fig. 6.

4.1. Medium duty vehicles (shuttles and minibus)

Three papers explored the security issues of shuttles which can be considered as medium duty vehicles. Authors of these publications did not concentrate on protocol or specific communication structure in shuttles or medium duty vehicles, they mostly concentrate on different consideration for these types of vehicles in terms of security concerns. Robert et al. explored the significance and knowledge gaps in the cybersecurity of autonomous vehicle (AV) shuttles. They developed an integrated cyber–physical testbed to evaluate the cybersecurity of these vehicles in real-world operations in Estonia. This testbed, designed to test and analyze the security features and vulnerabilities of automotive systems, is a controlled and simulated environment. In addition, they tested eight different attack scenarios, including denial-of-service (DoS) attacks and spoofing. They analyzed the consequences of these attacks using their testbeds, defining the boundaries and critical points of cybersecurity for real-world applications (Roberts et al., 2021). Benyahya et al. in 2022, conduct a comprehensive review and classification of surface attacks on Automated City Shuttles, detailing the vulnerabilities arising from their heterogeneous nature. They map various attacks to corresponding mitigation strategies, recommending a combination of countermeasures tailored to each attack type, including an overview of the advantages and disadvantages of each scheme. Their cybersecurity layers are divided into two main groups: in-vehicle and communication or external vehicles. Each attack surface is discussed separately, covering elements like Lidar, radar, and CAN for in-vehicle attacks, and Vehicle-to-Infrastructure (V2I) and Vehicle-to-Vehicle (V2V) for external communication. Additionally, they explore cybersecurity regulations for vehicles, categorizing them based on different regions worldwide (Benyahya et al., 2022). Benyahya et al. in 2023 again conducted a comprehensive study to assess risks associated with automated city shuttles at level 4 automation using Threat Analysis and Risk Assessment (TARA). They established their framework based on the ISO/SAE 21434 TARA process (Benyahya et al., 2023). The overview of the reviewed papers on MD vehicles is illustrated in Table 3.

4.2. Industrial HD vehicles

In this study, industrial HD vehicles refer to any vehicles used for industrial activities, such as agricultural and mining vehicles. The existing literature primarily discusses the importance of security and cybersecurity for these types of vehicles through various threat examples. Sontowski et al. underscore the need for robust security measures in smart farming due to the increasing reliance on interconnected devices and systems. These systems, which include IoT devices, sensors, and automated machinery, are vulnerable to cyber-attacks that could disrupt agricultural operations and compromise data integrity. They address the critical importance of security in smart farming, highlighting various threats such as password cracking, Evil Twin access points, and key reinstallation attacks. Their study emphasizes the role of different smart devices, including agricultural vehicles, but does not directly focus on the security of embedded systems or vehicular systems, though it mentions some attacks threatening vehicular systems (Sontowski et al., 2020). Gaber et al. (2021) conducted a comprehensive survey on Autonomous Haulage Systems (AHS), focusing on their safety, cybersecurity, and communication aspects. Their study provided a comparative analysis of these systems. They indicated that AHS are designed to mitigate the vulnerabilities and risks associated with manual vehicles that interact with auxiliary equipment. These systems utilize wireless communication technologies, including object avoidance/detection systems, Global Positioning Systems (GPS), GNSS, and artificial intelligence. Given the critical role of communication structures in AHS, the authors emphasize the necessity of securing communication systems to ensure safer, more productive, and sustainable operations. Insecure communication can lead to equipment damage, production losses, severe injuries, and fatalities. However, the study did not delve into specific attack surfaces, cyber-attacks, risk assessments, or mitigation strategies. Instead, it primarily focused on the relationship between the safety of AHS and the importance of secure communication and cybersecurity (Gaber et al., 2021).

Among the extracted publications exploring industrial vehicles, only one investigates protocol and embedded systems. Hariharan et al. conducted a study on the security of construction machinery, emphasizing the importance of safeguarding these vehicles against potential attacks by identifying vulnerabilities that attackers could exploit. They focused on the security testing of the SAE J1939 protocol, performing fuzz testing and penetration testing (pen testing) using a real ECU and an intruder device connected to the same ECU. Although their research primarily targets J1939, which is commonly used in heavy-duty vehicles, their findings are not limited to construction machinery but are applicable to a broader range of heavy-duty vehicles (Hariharan et al., 2022).

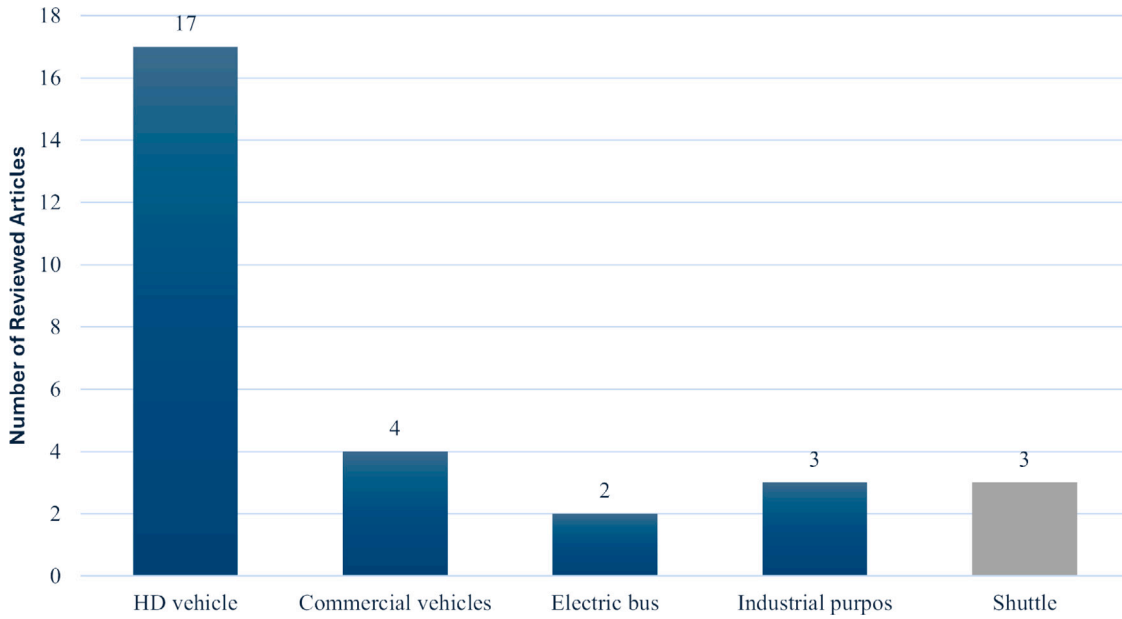

Fig. 6. Overview of Integrating Papers with Vehicle Types.

Table 3
MD vehicles review papers.

No	Ref.	Year	Cyber-Attacks detection	EV	AV	TARA	Novelty
1	Roberts et al. (2021)	2021	Different attack scenarios, including denial-of-service (DoS) attacks and spoofing.	Yes	Yes	No	Designing test bed
2	Benyahya et al. (2022)	2022	In-vehicle communication and external vehicle attacks.	No	Yes	No	Exploring attack surface
3	Benyahya et al. (2023)	2023	STRIDE for each detected asset.	No	Yes	Yes	TARA implementation

4.3. Electric buses

Two publications have been extracted exploring the security and cybersecurity of electric buses, concentrating on battery management systems and vulnerabilities associated with V2X communication. However, these publications did not discuss the specific protocols of HD or MD vehicles. Khani et al. conducted a study to explore cyber-threats in electric mobility systems (EMS), focusing exclusively on battery-based electric bus (BEB) fleets. This paper identifies and addresses cybersecurity threats to BEB fleets by examining their vulnerabilities to cyberattacks. The researchers employ an optimization model for scheduling battery swaps in the buses and use this model to assess the system's security. They introduce novel methods for detecting false data injection attacks (FDIAs), which involve inserting incorrect data into the system to disrupt its operations. To identify potential cyberattacks, the researchers utilize a supervised machine learning method. They integrate this cybersecurity model with the BEB scheduling model to develop a system capable of withstanding cyberattacks. In short, the primary focus of the study is on the battery system rather than the protocols or communication structures in heavy-duty vehicles. This concentrated approach allows for a detailed analysis of the specific vulnerabilities and threats related to the battery systems in BEB fleets (Khani et al., 2024). Shi et al. conducted a study exploring the integration of cloud computing in intelligent transportation systems to enhance the performance of electric city buses (ECBs). They focused on the cybersecurity challenges posed by Vehicle-to-Cloud (V2C) connectivity, particularly addressing denial-of-service (DDoS) attacks. Their proposed security architecture for ECBs includes detection and protection systems designed to mitigate these attacks. Their findings indicate that distributed DDoS attacks are especially critical due to their ease of execution and difficulty in defense. While this study did not concentrate on the specific protocols of HD vehicles, it emphasized the wide range of communications of electric buses and V2X, which give rise to more cyber-attacks for these kinds of vehicles compared to regular vehicles (Shi et al., 2023). The overview of the reviewed papers on electric buses is illustrated in Table 4.

4.4. Commercial vehicles and HD vehicles

Most of the publications extracted in this study have been categorized as HD vehicles in general. In fact, these publications explored HD vehicles more comprehensively while some articles named them commercial vehicles and some Heavy-duty vehicles. More exploration in terms of different protocols and structure of HD vehicles can be found in these publications. In the following, different focus areas extracted from publications are discussed.

4.4.1. Protocol J1708/J587

Nnaji and Daily conducted a research project focusing on the security of the connection between tractor and trailer in HD vehicles, noting that these connections have not been sufficiently addressed. To enhance security, they proposed the Legacy Intrusion Detection System

Table 4
Electric buses review papers.

No	Ref.	Year	Attack surface explored	EV	AV	TARA	Novelty
1	Khani et al. (2024)	2024	Battery swapping station	Yes	No	No	Exploring battery management system for electric buses
2	Shi et al. (2023)	2024	Connectivity to cloud	Yes	No	No	Exploring the vulnerabilities of electric buses in terms of communication and connectivity

(LIDS) for vehicles using the SAE J1708/J1587 protocol. The authors identified several attack types, including message spoofing, flooding, attacks from hosts, and flooding attacks from rogue nodes. To protect against these attacks, LIDS employs gateways at each host to monitor network traffic using specialized J1587 security messages. The system utilizes both signature-based and anomaly-based detection methods to identify threats, relying on configurable parameters like busload, access control, and transmission rates to perform detection on inbound and outbound network traffic (Nnaji and Daily, 2024).

4.4.2. Protocol J1939

In this group of publications, there is a more detailed examination of J1939 and the data transfer processes within the J1939 framework. These articles primarily approach J1939 as a significant attack surface rather than a "black box". To organize these works, we have categorized them by their focus areas. The first category includes publications that address vulnerabilities and potential threats targeting J1939, along with mitigation strategies and recommendations. The second category focuses more on cybersecurity methods tailored to J1939 than on exploring specific threats.

4.4.2.1 Threat and attack discussions

General analysis of J1939 vulnerabilities These publications examine the weaknesses and vulnerabilities inherent in J1939, identifying areas that attackers may exploit. One of the most significant studies in this area was conducted by Burakova et al. where they highlight that attackers can gain physical access to heavy-duty vehicles via the OBD port to transmit arbitrary messages. They also identified key vulnerabilities, including the connection between the tractor and trailer, and fleet management systems commonly used in heavy-duty vehicles, which create additional attack surfaces. Burakova et al. emphasize that J1939 is an open standard, meaning that an attacker with basic technical expertise and sufficient resources could exploit its vulnerabilities. In their experiment on a Class 8 truck, they extracted data from the OBD port and injected malicious messages into J1939. Their findings demonstrated that an attacker with access to the J1939 protocol could manipulate safety-critical systems in heavy-duty vehicles. They specifically tested attacks on the instrument cluster, powertrain, and engine brake, highlighting the serious risks posed by such vulnerabilities (Burakova et al., 2016).

Murvay and Groza expand on the shortcomings of J1939 and outline several potential issues that attackers could exploit to carry out Denial of Service (DoS) attacks. They identify key weaknesses, starting with the address claim process, where attackers can pose as legitimate nodes and claim the same address by using a lower "NAME" field value, thereby blocking the real node from communicating. Additionally, network flooding attacks can overwhelm the system by sending excessive data requests, delaying or blocking critical messages. Lastly, during large data packet transmissions, attackers can send fake abort messages, prematurely halting communication between nodes and disrupting data transfer. To mitigate these risks, Murvay and Groza propose the use of Public Key Infrastructure (PKI) as an effective security measure for J1939 systems. They argue that PKI enables secure communication without relying on secret keys, which are difficult to manage and exchange securely over the CAN bus. They validate their PKI solution in a simulated environment using CANoe, which is a tool for testing CAN-based networks (Murvay and Groza, 2018).

Investigating specific attacks on J1939 protocol These articles explores and discusses specific attacks targeting the J1939 protocol. Mukherjee et al. focused their research on identifying weaknesses in the data link layer of the SAE J1939 protocol, specifically examining Denial-of-Service (DoS) attacks in commercial vehicles. They highlighted that protecting vehicles from such attacks requires a detailed understanding of the J1939 protocol workflow to identify exploitable interfaces. The authors analyzed the structure of J1939 frames and how communication and data transfer take place within the system. They categorized DoS attacks into three main types: request overload, false request to send (RTS), and connection exhaustion. In a request overload attack, an attacker overwhelms the ECU with excessive request messages, causing it to fail in performing critical functions. In RTS attack, spoofed RTS messages with altered data sizes can lead to buffer overflow and ECU firmware crashes. Finally, connection exhaustion involves an attacker opening multiple simultaneous connections to a critical ECU, such as a brake controller, preventing other ECUs from establishing connections. The authors also explored mitigation strategies for each of these attacks (Mukherjee et al., 2016). Similarly, Chatterjee et al. building on previous works on J1939 OSI layer attacks, investigated data link layer vulnerabilities in J1939, examining two methods for data transmission: Destination-Specific Transfer and Broadcast Transfer. They conducted experiments with five distinct attacks across four testbeds with various ECM (Engine Control Module) and EBC (Electronic Brake Controller) setups. These attacks included request overload, where excessive directed request messages cause the ECU to neglect critical tasks, mitigated by rate-limiting incoming requests; connection exhaustion, where fake connections via CTS messages block legitimate access, mitigated by automatic termination of inactive connections; BAM blocking, where fake destination-specific requests disrupt global broadcasts, mitigated by ensuring BAMs for specific PGNs are always transmitted; malicious CTS, where invalid packet requests create confusion, mitigated by ignoring such requests; and memory leaks, where invalid CTS messages cause unintended data transmission, also mitigated by discarding invalid messages. These findings highlight the importance of understanding and addressing J1939 protocol vulnerabilities to safeguard critical vehicular systems (Chatterjee et al., 2023).

Jichici et al. defined the structure of the CAN bus protocol, with significant emphasis on the J1939 frame and data transmission methods between ECUs. They conducted an experimental study to detect and evaluate attacks on J1939, concluding that machine learning algorithms are not sufficient for detecting such attacks due to the lack of available time for training. Utilizing Simulink and CANoe, they simulated the J1939 structure to evaluate threats to HD vehicles. They measured vehicle speed, trip distance, and engine speed under four different kinds of attacks: (a) fuzzing attack, (b) surge attack, (c) bias attack, and (d) geometric attack (Jichici et al., 2023).

4.4.2.2 Cybersecurity methods and mitigation strategies for J1939

Anomaly detection Shirazi et al. conducted a research utilizing machine learning techniques to detect anomalies and malicious activities in J1939 data transmissions. Focusing on HD vehicles employing the J1939 protocol, they highlighted the heightened vulnerability of such vehicles due to the open-source nature of the protocol. They emphasized that deciphering the coding in J1939 is relatively easy for attackers, making protection of data transmission in J1939 imperative, despite the limited number of research projects addressing its security. The researchers proposed employing machine learning algorithms to detect anomalies, malicious activities, and abnormal traffic exclusively

in the embedded network of J1939 in HD vehicles. They argued that the complexity of data transmission in J1939, owing to its structured data transfer format, poses significant challenges for abnormal traffic detection, necessitating advanced methodologies such as machine learning. Their learning algorithm creates a behavioral profile for each Parameter Group Number (PGN) and employs machine learning techniques to detect abnormal messages (Shirazi et al., 2020). Mukherjee et al. conducted additional research addressing the vulnerability of embedded devices in commercial vehicle networks to message injection attacks, leveraging the widely adopted J1939 standards. These standards simplify the identification of command messages but also expose vehicles to potential exploitation through spoofed messages, leading to unintended actions by ECUs. The study introduces a novel precedence graph-based anomaly detection technique designed to detect and mitigate such malicious injections and associated probing or fuzzing activities. This approach aims to safeguard vehicular networks against attacks that mimic critical safety actions, thereby enhancing network security despite the inherent openness of J1939 standards and obscured OEM-specific engineering details (Mukherjee et al., 2017).

Intrusion detection Jichici et al. proposed an intrusion detection and prevention system for the J1939 protocol, emphasizing that traditional intrusion detection methods for CAN often treat it as a "black box" and need to be expanded to address the complexities of J1939. To support their work, they built a real test bed for a HD vehicle and collected data directly from the J1939 diagnostic port. They focused on detecting replay and modification attacks. In replay attacks, where attackers retransmit captured data to deceive the system, the use of encrypted addresses helps detect and block these attempts. In modification attacks, where the transmitted data is altered, range checks and the avalanche effect in encrypted data fields are used to detect tampering, as even small changes in the input cause significant output differences. Their proposed two-layer intrusion detection system includes encrypted addresses, where AES encryption replaces regular sender and receiver node addresses, making it harder for attackers to reuse known addresses, and data-field encryption and authentication, which adds a layer of protection by concealing the true content of the CAN frame with lightweight encryption methods (Jichici et al., 2022).

Authentication Palaniswamy et al. conducted a comprehensive study on enhancing authentication protocols for intra-vehicular networks using the SAE J1939 standard, which is crucial for securing ECUs in commercial vehicles. Their research addresses vulnerabilities such as replay, masquerading, and machine-in-the-middle attacks within the standard, emphasizing the critical need for robust security measures. The study includes a formal analysis of existing protocol suites and introduces two innovative authentication protocols: a computationally efficient one-pass protocol tailored for restricted nodes and a certificateless signature-based protocol suitable for other nodes. Additionally, they propose a novel certificateless key insulated manageable signature (CL-KIMS) scheme designed to enhance security by ensuring key insulation, facilitating random access key updates, and providing self-healing capabilities. Through rigorous security assessments and performance evaluations, the authors demonstrate the efficacy of their protocols in mitigating attacks and minimizing communication overhead compared to current solutions. Their findings contribute significantly to advancing the security of commercial vehicle networks against evolving cyber threats (Palaniswamy et al., 2022).

Encrypting Diagnostic Interfaces Encryption: In an experiment, Daily and Kulkarni explore the security of diagnostic connections, specifically focusing on the RP1210 protocol. They highlight that heavy vehicles require regular maintenance, during which a technician connects a diagnostic tool (Vehicle Diagnostic Adapter, or VDA) to the vehicle's diagnostic port and runs diagnostic software on a computer to check the vehicle's condition. Since this process happens frequently, it poses a potential security risk, especially when the diagnostic tool is connected to the internet. To address this, the authors propose encrypting the communication between the diagnostic application and the vehicle's secure gateway using a strong encryption method, AES-128, which ensures that messages are protected from unauthorized access or modification. For their cybersecurity experiment, they examine two types of "man-in-the-middle" attacks: single-frame and multi-frame attacks (Daily and Kulkarni, 2020).

Bit-Banged J1939 filtering Campo et al. highlighted the increasing connectivity of heavy-duty (HD) vehicles due to the integration of wireless telematics devices and the rapid digitalization in this field, underscoring the critical importance of security in HD vehicles. They examined the differences between the CAN and J1939 protocols, focusing specifically on the address claim attack within J1939. Unlike CAN, J1939 requires a node to claim an address before transmitting or receiving data on the network. In this process, the controller sends an address claim message and awaits any contending claims. Given J1939's arbitration system, nodes assign priority to claims with lower NAME values, meaning a lower NAME claim holds higher priority. An address claim attack occurs when an attacker transmits a claim message with a lower NAME value, thus gaining higher priority on the network. To counter this attack, they propose a device called ClaimDefender, which is bit-banged CAN filter and does not rely on a central host-based system for security measures (Campo et al., 2021).

4.4.3. CAN Bus

Shipman et al. concentrating on intra-vehicle networks, delve into the implementation of an automotive Zero Trust Architecture (ZTA) to fortify cybersecurity within commercial vehicles, which have increasingly become targets of cyberattacks since the 1990s. Zero Trust (ZT), originating from Information Technology (IT), underscores the principle of never assuming trust in any asset, irrespective of its ownership or network location. By scrutinizing threats like illicit message injection, unauthorized firmware installation, message spoofing, and unauthorized access, the research endeavors to markedly enhance the security of the vehicle's cyber–physical network. Concurrently, the aim is to minimize adverse impacts on performance, such as timing, bandwidth, and processing power. While they address concerns about commercial vehicles, they do not explore J1939, and their results are applicable to regular vehicles as well (Shipman et al., 2024).

4.4.4. External connection

As previously mentioned, external connections serve as a primary attack surface and warrant further attention. As discussed in Section 3.2, attackers can exploit various connections in HD vehicles. Consequently, the articles are categorized based on the specific connections they explore.

Interfaces and connections for diagnostics and monitoring Chatterjee et al. focus on the inherent vulnerabilities of UDS (Unified Diagnostic Service) in heavy-duty (HD) and medium-duty (MD) vehicles. They highlight that while UDS and diagnostic ports and protocols play a crucial role in vehicle diagnostics, programming, and fault resolution, their inherent vulnerabilities need to be addressed. As part of their evaluation, they explored the Freightliner Cascadia cab setup. They indicate that UDS can resist spoofing attacks, but almost all types of diagnostic messages can penetrate UDS, making it vulnerable to diagnostic-based attacks. To assess UDS vulnerabilities, they conducted experiments on four different testbeds involving various component combinations. Their experiments revealed three different scenarios. The first scenario is the Read Data by ID Overload Vulnerability, which occurs when ECUs (Electronic Control Units) are overwhelmed by excessive "Read Data by ID" requests. To mitigate this vulnerability, it is essential for ECUs to ignore such requests if received faster than a predetermined threshold. The second scenario is the Session Denial Vulnerability, where a single session request can lock an ECU, preventing it from responding to other diagnostic requests. Implementing

Table 5
Overview of reviewed papers on HD vehicles.

No.	Ref.	Year	J1939	AV	TARA	Attack surface	Explored attacks	Security countermeasure
1	Nnaji and Daily (2024)	2024	No	No	No	Connection between tractor and trailer	Flooding Attacks	Intrusion Detection System (LIDS)
2	Shirazi et al. (2020)	2020	Yes	No	No	J1939	Manipulating data in PGN in J1939	Anomaly detection (via machine learning)
3	Wolf and Lambert (2017)	2017	No	No	Yes	General attack surface	Physical theft, data attack, attack on vehicle safety and reliability	TARA implementation and security protection for system, lifecycle, and organization
4	Mukherjee et al. (2017)	2018	Yes	No	No	J1939	Probing, fuzzing, spoofing	Anomaly detection (via graph-based method)
5	Elhadeedy and Daily (2023)	2023	No	Yes	No	Connection between tractor and trailer	Wifi crash, sniffing, rogue Wi-Fi access points, unauthorized client, replaying modified or old data, Wi-Fi credentials exposed, Wi-Fi interference, jamming	Using Named Data Networking (NDN) to secure in-vehicle and cloud communication
6	Chiocchio et al. (2018)	2018	No	No	No	Communication (V2X)	Jamming, spoofing	Some explorations to secure communication and accurate geo-localization
7	Ahmed (2024)	2024	No	No	Yes	General attack surface	Replay attacks and malware attacks	TARA implementation
8	Shipman et al. (2024)	2024	No	No	No	Intra-vehicle networks	Illicit message injection, unauthorized firmware installation, message spoofing, and unauthorized access	Zero Trust Architecture (ZTA) to secure intra-vehicle networks
9	Palaniswamy et al. (2022)	2023	No	No	No	J1939	Masquerading, machine in the middle, man in the middle	A certificateless signature-based authentication protocol and certificateless key insulated manageable signature (CL-KIMS)
10	Jichici et al. (2023)	2024	Yes	No	No	J1939	Reply attack, DoS attack, fuzzing attack	Countermeasures based on experiments
11	Campo et al. (2021)	2021	Yes	No	No	J1939	Address claim attack	Using bit-banged Controller Area Network (CAN) filter to detect the attack
12	Kumar et al. (2023)	2023	No	No	No	Diagnostic ports and J1939	Machine in the middle	Machine authentication codes or authenticated encryption with pre-shared keys
13	Daily et al. (2016)	2016	No	No	No	Intra-vehicle networks		Designing testbed for HD vehicles cyber-attack
14	Chatterjee et al. (2024)	2024	Yes	No	No	UDS and J1939	1.Read Data by ID Overload Vulnerability, 2.Session Denial Vulnerability, 3.Diagnostics Jam Vulnerability	1.Ignore rapid request 2.Session request queue 3.Timeout adjustments and anomaly detection
15	Jepson et al. (2024b)	2024	Yes	No	No	ELD and J1939	Truck to truck cyber worm attacks	strengthening default security settings, integrity and authenticity checks for firmware, implementing telematic device firewalls or gateways
16	Murvay and Groza (2018)	2018	Yes	No	No	J1939	Address claim process issues, network flooding attacks, and exploitable connection modes for larger data packets disrupting communication	Public Key Infrastructure (PKI)
17	Mukherjee et al. (2016)	2016	Yes	No	No	J1939	Dos attack under subcategories: Request overload, False RTS and connection exhaustion	Programming ECU to ignore repeated request, Using dynamic memory allocation, Authenticating the sending ECU
18	Jichici et al. (2022)	2022	Yes	No	No	J1939	replay and modification attacks	encrypted addresses data-field encryption
19	Burakova et al. (2016)	2016	Yes	No	No	J1939	safety-critical attacks	Incorporate active adversary considerations in safety feature design
20	Chatterjee et al. (2023)	2023	Yes	No	No	J1939	Request Overload, Connection Exhaustion, BAM Block, Malicious CTS, Memory Leak	Rate-limiting requests, Auto-terminate inactive connections, Prioritize BAMs, Ignore invalid CTS, Discard invalid CTS
21	Daily and Kulkarni (2020)	2020	Yes	No	No	J1939, Diagnostic interfaces	Man-in-the-middle	Encryption Method, AES-128

a session request queue system can help the ECU manage multiple session requests efficiently, rather than being locked into handling just one. Lastly, the Diagnostics Jam Vulnerability arises from abnormal flow control conditions disrupting normal communication between the ECU and diagnostic tools. Mitigation strategies for this vulnerability include adjusting protocol timeouts to counter abnormal conditions, using anomaly detection algorithms to identify and react to suspicious patterns of Flow Control (FC) frames, and enhancing the ECU's capabilities to better manage high volumes of these frames (Chatterjee et al., 2024).

Jepson et al. explore the security vulnerabilities of Electronic Logging Devices (ELDs), which are commonly used in commercial vehicles by connecting to J1939, OBD, or RP1226 networks. ELDs monitor the intra-vehicle network and extract critical data such as engine hours, vehicle motion, and distance traveled. After introducing ELDs, they provide a comprehensive threat model, highlighting how an attacker could take control of an ELD, either remotely or physically, inject malicious firmware, and then infect nearby devices with malware that waits for an opportune moment to cause dangerous malfunctions. The researchers examined three key interfaces—GPS, WiFi, and Bluetooth—through reverse engineering and evaluation of default and factory settings, uncovering vulnerabilities. While their focus was on identifying potential threats, they also proposed cybersecurity countermeasures. These include strengthening default security settings, establishing strong integrity and authenticity checks for firmware, implementing telematic device firewalls or gateways, and removing unnecessary or high-risk features to mitigate security risks (Jepson et al., 2024b).

Kumar et al. studied a specific attack in HD vehicles known as Machine-In-The-Middle (MitM). Previously, it was assumed that MitM attacks involved altering hardware to manipulate CAN bus traffic. However, their study shows that MitM attacks can occur even without direct tampering with the CAN bus. Instead, these attacks exploit vulnerabilities in diagnostic applications based on RP1210, (a protocol standard developed by the Truck and Bus Control and Communications Network (TMC) for vehicle diagnostics). Their test results demonstrated that SAE J1939 communications are vulnerable to data manipulation attacks resulting from MitM, regardless of whether the data is sent in single-frame or multi-frame messages. They concluded that by using machine authentication codes or authenticated encryption with pre-shared keys as a solution, it becomes much harder for attackers to intercept or manipulate the messages without detection, thus mitigating the risk of MitM attacks (Kumar et al., 2023).

Tractor-Trailer connections Elhadeedy and Daily conducted a study on securing the communication between tractor and trailer components in trucks. These trucks consist of two separate parts—the tractor and the trailer—that are connected via interfaces such as power line communication (PLC) and ISO11992-CAN. However, there is a growing need for autonomous coupling and uncoupling of the trailer from the tractor. For this autonomous process, a higher bandwidth connection is necessary, so they proposed utilizing a wireless harness to connect the trailer's ABS to the tractor's ECU. They argued that the existing networks employed in trucks, such as J1939, are insufficient for the communication requirements of ECUs in autonomous trucks. Instead, they suggested the use of networks like Ethernet and CAN FD. While these proposed communications enhance the capabilities of autonomous trucks, they also introduce more vulnerabilities. To address these vulnerabilities, Elhadeedy and Daily emphasized the importance of authenticating each part before accessing the wireless network to ensure secure pairing. They outlined four scenarios for connection and authentication processes and identified seven different risks associated with the connection between tractor and trailer. These risks include WiFi crashes, WiFi spoofing, rogue WiFi access, WiFi jamming, exposed WiFi credentials, replaying modified or outdated data, and unauthorized clients. For each risk, they discussed the cause, risk level, failure mode, and proposed countermeasures (Elhadeedy and Daily, 2023).

Vehicle to everything (V2X) connections Chiocchio et al. conducted research focusing on vehicular connectivity, specifically addressing the navigation of commercial vehicles. Their study explores techniques and technologies for ensuring secure communication and accurate geo-localization, with a particular emphasis on vehicles equipped with fleet management systems. It focuses on Intelligent Transportation Systems (ITS) within vehicles, defining and exploring Vehicle-to-Everything (V2X) communication. The study outlines various use cases associated with ITS, including cooperative road safety and traffic efficiency. Additionally, it examines satellite systems for navigation and localization algorithms. The study also addresses the diverse threats to Global Navigation Satellite Systems (GNSS) receivers, which play a pivotal role in determining a vehicle's location based on signals received from satellites. The threats are categorized as follows: Jammers are devices used for denial-of-service attacks by emitting interference signals that block GNSS receivers from accurately determining their location. Spoofers and meaconers pose a more severe threat by intentionally manipulating GNSS signals to deceive receivers into calculating false positions or times, which can lead to serious risks in critical applications like navigation and timing synchronization (Chiocchio et al., 2018).

4.4.5. TARA

Threat Analysis and Risk Assessment (TARA) is a systematic process used to identify potential threats and evaluate the risks they pose. It helps prioritize security measures based on the likelihood and impact of these threats. In the reviewed publications, two papers studied TARA, which have been discussed in the following. Wolf and Lambert (2017) conducted a comprehensive exploration to implement TARA for HD vehicles. In the first section of their publication, they explored the importance of HD vehicles in terms of security and studied the potential threats in general. They identified four main threats for HD vehicles, including physical theft of complete vehicles or valuable vehicle components, data theft attacks or misuse of digital vehicle data, and attacks on vehicle reliability and vehicle safety. They compared each threat between HD vehicles and passenger cars in terms of attack probability, damage potential, damaged party, and typical attacker. After assessing the level of attack success probability and damage potential, they analyzed these four threats based on the TARA matrix and level of security. Finally, they provided three main protection approaches for HD vehicles: security for the entire heavy-duty vehicle system, security for the entire heavy-duty vehicle lifecycle, and security for the entire heavy-duty vehicle organization. To the best of our knowledge, their study was the first TARA exploration for HD vehicles, although they only explored generic threats without expanding on protocols and HD vehicle structure details (Wolf and Lambert, 2017).

Ahmed in 2024. proposed an automated TARA model for heavy-duty (HD) vehicles based on ISO/SAE 21434. They emphasized the importance of TARA and introduced each section, including risk assessment and threat analysis, along with their parameters and steps in ISO/SAE 21434. Leveraging the Microsoft Threat Modeling Tool (MSTM) and data flow diagrams (DFDs), they evaluated two main attacks: replay attacks and malware attacks. While their TARA model focused on HD vehicles with J1939, they highlighted that their approach and model are practical for all types of vehicles, not just HD vehicles (Ahmed, 2024).

4.4.6. Testbed

A testbed is an experimental platform used for testing, simulating, and evaluating technologies in controlled environments. It is crucial for research and development, particularly for assessing the security and performance of systems like HD vehicles. By explaining the differences between HD vehicles and LD ones, Daily et al. in 2016 designed and provided a customized testbed for cyber-attacks on HD vehicles.

Testbeds are crucial in the research of HD vehicle security because they allow for the detection, experimentation, and evaluation of both physical and cyber aspects of attacks on these vehicles, which are typically expensive to acquire and maintain. The authors believe that the most important aspect of a testbed is its impact on network communication, rather than sensor simulation and determinism. However, they did not focus on a specific type of attack or cybersecurity threat, although they did provide some features and characteristics of HD vehicles that were used in designing the testbed (Daily et al., 2016).

An overview of the reviewed papers exploring J1939 is presented in Table 5.

5. Identifying research gaps

The current literature on HD vehicle cybersecurity primarily addresses general HD vehicles, focusing on J1939 protocol vulnerabilities. However, there remains a significant research gap in examining the cybersecurity needs of specialized HD vehicles, such as military, emergency, and industrial vehicles (e.g., those used in construction and agriculture). These vehicles often feature unique interfaces and communication protocols tailored to their specific applications, necessitating a more comprehensive approach to cybersecurity.

Furthermore, the rising adoption of electric HD vehicles, particularly electric city buses, has not been thoroughly investigated in terms of cybersecurity, leaving critical gaps in securing these emerging technologies. Additionally, as autonomous and intelligent vehicle technologies evolve—especially in driverless vehicles for industrial and special purposes—their unique vulnerabilities demand closer examination. Autonomous HD vehicles, in particular, are often more susceptible to cyber threats due to their reliance on complex software and networked systems.

Moreover, there is a notable lack of research focusing on Threat Analysis and Risk Assessment (TARA) methodologies within the HD vehicle sector. Research on external communication interfaces, such as Vehicle-to-Vehicle (V2V) and Vehicle-to-Everything (V2X) communications, is also limited, despite the growing importance of these connections. Finally, diagnostic connectors, including OBD and ELD devices, present additional potential vulnerabilities that warrant further investigation. Addressing these gaps is essential to advance the cybersecurity posture of HD vehicles in both current and emerging technological landscapes.

6. Conclusion

This article analyzes the cybersecurity differences between light-duty and heavy-duty vehicles, revealing significant disparities. The review of relevant publications highlights the increased susceptibility of HD vehicles to cyber threats, attributed to factors such as the use of open-source communication protocols like J1939, complex system architectures, numerous physical interfaces, and extensive fleet communication systems. In comparison to standard vehicles, HD vehicles face heightened risks. Furthermore, the literature review uncovers a notable research gap in addressing these critical vulnerabilities. Few studies specifically examine J1939 and intra-vehicle network security, despite their crucial role in vehicle operations. This lack of comprehensive research emphasizes the urgent need for more focused investigations into HD vehicle cybersecurity. Moving forward, it is imperative to concentrate on understanding and mitigating these vulnerabilities to protect HD vehicles from the increasing array of cybersecurity threats. To safeguard the resilience and security of commercial vehicle fleets and public transportation systems, it is essential for researchers and industry stakeholders to focus on this area.

CRediT authorship contribution statement

Narges Rahimi: Writing – original draft, Methodology, Investigation, Formal analysis. **Beth-Anne Schuelke-Leech:** Writing – review & editing, Validation, Supervision. **Mitra Mirhassani:** Supervision, Resources, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Narges Rahimi reports financial support was provided by Mitacs Canada. Narges Rahimi reports financial support was provided by Federal Economic Development Agency for Southern Ontario. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Narges Rahimi is a Ph.D. candidate in the Department of Mechanical, Automotive, and Materials Engineering at the University of Windsor, Windsor, ON, Canada. She received her Master of Science in Industrial Engineering from Iran University of Science and Technology, Tehran, Iran, in 2016, and her Bachelor of Science in Electrical Engineering from Shahid Beheshti University, Tehran, Iran, in 2013. Her research focuses on the security and cybersecurity of vehicles, particularly heavy-duty vehicles, by exploring standards and frameworks. She specializes in manufacturing and industrial system engineering, with a strong emphasis on cybersecurity and risk assessment.

Beth-Anne Schuelke-Leech is an Associate Professor of Engineering Management at the University of Windsor in Canada. She earned an undergraduate degree in Mechanical Engineering from McMaster University in 1994, an MBA from York University in 2004, and a Ph.D. in Public Policy at the University of Georgia in 2011. She was a faculty member at the John Glenn School of Public Affairs at the Ohio State University from 2011 to 2016. In 2016, she joined the Faculty of Engineering at the University of Windsor. Before undertaking her doctoral studies, she worked for 12 years as a professional engineer, primarily in alternative fuel product development for General Motors of Canada. Dr. Beth-Anne Schuelke-Leech is an interdisciplinary scholar whose work focuses on the intersection of technology development with society and the impacts of disruptions. She is currently engaged in projects encompassing cybersecurity, ethics, public policy, electric battery design, the circular economy, smart cities, and enhancing resilience in systems.

Mitra Mirhassani is a professor at the Department of Electrical and Computer Engineering, University of Windsor. Her research focuses on hardware security, the development of methods for the detection of Trojan Hardware, and high-performance custom-designed AI systems. She is a member of the APMA Cyber Security Committee since 2019 and currently serves as the Co-Chair of the committee. She served as Vice-Chair for Women in Engineering Affinity Group, IEEE South-East Michigan section (2014), and an advisor to WiCyS, Windsor Chapter (2020). She is serving as the Associate Editor for IEEE Access Journal and Neural Network track chair for the 2016–2018 IEEE International Symposium on Circuits and Systems. She is an IEEE Conference Editorial Board Member and a Senior Member of IEEE. She has been recognized by the IT world Canada as one of the top women in cybersecurity in Canada in 2020 and won the 2020 APMA's cyber Security Individual Achievement Award. She received the 2021 APMA leadership awards in cybersecurity.